

Trends in death certification for multiple sclerosis, motor neuron disease, Parkinson's disease and epilepsy in English populations 1979–2006

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Abstract The objective of this study is to report trends in mortality, as certified on death certificates, from multiple sclerosis (MS), motor neuron disease (MND), primary Parkinson's disease (PD), and epilepsy, analysing not only the underlying cause of death but also all certified causes for each disease. Death records in the Oxford region, 1979–2006, and England, 1996–2006, were analysed for ascertaining the trends in mortality. The percentage of deaths coded as the underlying cause changed over time, coinciding with changes to the rules for selecting the underlying cause of death. Changes over time to coding rules had a large impact on apparent trends in death rates for PD when studied by underlying cause alone. They also had significant, though smaller, effects on trends in death rates for MS, MND and epilepsy. Nationally, in the last period of the study, underlying cause mortality identified 64% of deaths with a mention of MS, 88% of MND, 56% of PD, and 48% of epilepsy. In the longstanding Oxford data from 1979 to 2006, death rates based on all certified causes of death showed no significant change for MS; an upward trend for MND (notably in women over 75), though only in the last few years of the study; a significant but small decline for PD; and no significant change for epilepsy. When mortality statistics are analysed by underlying cause only, their value is reduced. A substantial percentage of

neurological deaths are missed. Time trends may be misleading. All certified causes for each disease, as well as the underlying cause, should be analysed.

Keywords Multiple sclerosis · Motor neuron disease · Parkinson's disease · Epilepsy · Mortality · Trends

Introduction

In most populations, mortality statistics are still the only readily available source of information about long-term trends in chronic neurological diseases. A limitation of routine mortality statistics is that, until recently in England and many other countries, only one underlying cause of death from each death certificate has been selected for routine coding and analysis. Typically, in chronic diseases, several co-existing conditions may contribute to death. When more than one disease contributes to death, statistics confined to only one cause for each death will lack precision and may lack credibility. Furthermore, any study of long-term trends in death rates is hampered, if confined to data on underlying cause alone, by the fact that the procedures for selecting the underlying cause change over time. For example, in England they changed in 1984, 1993 and 2001 [19–21]. These changes had an impact on mortality statistics for neurological conditions [11, 24].

In the former Oxford National Health Service region of England, all causes of death on death certificates, contributing causes as well as the underlying cause, have been recorded since 1979. Nationally, such recording began for England in 1993 (although we only have access to full national data from 1996). In the present study, we analysed death rates for multiple sclerosis (MS), motor neuron

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disease (MND), Parkinson's disease (PD) and epilepsy, in the regional and national populations, using all certified causes of death (conventionally termed 'mentions' of causes of death). Our aims were to determine how death rates have changed over time and to identify potential sources of artefact in analyses based on underlying cause alone.

Method

The regional data were collected in the former Oxford National Health Service region, which covered a total population of about 2.5 million people, for the Oxford record linkage study [11]. The population comprises the counties of Oxfordshire, Berkshire, Buckinghamshire and Northamptonshire. Data were coded according to the prevailing English national coding conventions at the time of death certification. The national data were from mortality files supplied to the South East Public Health Observatory by the Office for National Statistics (ONS) for 1996–2006. In both sets, deaths were coded to the ninth revision of the International Classification of Diseases (ICD9) [31] for the years prior to 2001 and to ICD10 [32] for 2001–2006.

The databases were searched for records of people with a death certificate that mentioned MS (ICD9 340; ICD10 G35), MND (ICD9 335.2; ICD10 G12.2), primary PD (ICD9 332.0; ICD10 G20) or epilepsy (ICD9 345; ICD10 G40–G41). We analysed the regional data in four main time periods defined by major changes to national coding rules, 1979–1983, 1984–1992, 1993–2000 and 2001–2006, to calculate age-specific rates for mentions and for underlying cause. Age-specific mortality rates in 5-year age-groups were used to calculate age-standardised mortality rates, with 95% confidence intervals, for each time period. We also calculated age-standardised rates for each individual calendar year. We standardised, using the direct method, by applying the age-specific rates in each 5-year age-group in each calendar year to the European standard population. We calculated the average annual percentage change across the whole period for mortality rates based on mentions, using data for each individual calendar year, by fitting linear regression models to the logarithms of the death rates. Changes between adjacent time periods were tested for significance using chi-square tests on the numbers of deaths certified as underlying cause, or certified elsewhere. To simplify visual presentation of the graphs we have presented confidence intervals for England, but not for the Oxford region (to avoid visual clutter) except for PD where numbers of deaths were sufficiently large that confidence intervals were small.

Results

Multiple sclerosis (Table 1; Fig. 1a)

In the Oxford region, death rates based on mentions in the four main periods of the study identified respectively 59% (152/258), 70% (387/552) and 52% (274/526) and 65% (279/428) of all deaths with a mention of MS (Table 1). The differences in percentages coded as underlying cause, between the first and second period, the second and third, and the third and fourth period were statistically significant. This means that death rates, based on underlying cause, apparently rise in the second period, fall in the third, and rise again from 2001. Death rates based on mentions showed little appreciable change over time: they were 32 per million in 1979–1983, 35 per million in 1984–1992, 30 per million in 1993–2000 and 30 per million in 2001–2006 (Table 1). The percentages of deaths coded as the underlying cause in the two periods covered by the English data (64 and 53%) were almost the same as those in Oxford.

Death rates were higher in women than men; they peaked in people aged 64–75 years (more strikingly so in men than women) and then tended to decline with increasing age (data not shown). There was some year-on-year fluctuation within the period 1979–2006 (Fig. 1a). Mortality rates in England were very similar to those in the Oxford region in the years for which there are comparable data (Table 1). In the much larger dataset for the whole of England, there was an evident stepwise increase in death rates for MS, as the underlying cause of death, from 1996–2000 to 2001–2006 (Fig. 1a). This coincides with the implementation of the 10th revision of the ICD, in 2001, and associated changes to selecting the underlying cause of death. There was no stepwise change in mortality rates over time based on mentions (Fig. 1a, upper broken line).

Motor neuron disease (Table 2; Fig. 1b)

Mortality based on underlying cause identified the great majority of deaths certified as MND. From 2001 to 2006 it identified 88.1% in Oxford and 87.6% in England. Mortality rates were higher in men than women and they increased with increasing age (data not shown). In the Oxford region from 1979 to 2006, the percentage of deaths with MND coded as the underlying cause was higher between 1984 and 1992 than in the adjacent periods. This means that MND death rates, if judged from underlying cause alone, would artefactually appear to be lower before 1984 than after (Table 2). The average annual age-standardised mortality rate for men, measured as underlying cause, was 31 in 1979–1983, 35 in 1984–1992, 30 in 1993–2000 and 33 in 2001–2006. Measured as mentions, the corresponding rates in these four time periods were 40, 38,

Table 1 Multiple sclerosis: age-standardised death rates in the Oxford Region 1979–2006 and in England 1996–2006: rates per million population; underlying cause deaths as a percentage of all mentions (UC/M %); average annual change in death rates based on mentions

Sex/Area	Number of deaths				Rates ^a per million							
	Mentions				Underlying cause				Mentions			
	1979–1983	1984–1992	1993–2000 ^b	2001–2006	1979–1983	1984–1992	1993–2000 ^b	2001–2006	1979–1983	1984–1992	1993–2000 ^b	2001–2006
Males												
Oxford	95	204	179	169	16.8	18.9	10.6	16.8	25.6	27.5	22.2	25.3
England	–	–	2,331	2,785	–	–	12.7	14.3	–	–	23.6	21.9
Females												
Oxford	163	348	347	259	21.8	31.3	21.8	23.0	38.9	42.9	38.6	35.1
England	–	–	4,197	5,422	–	–	20.3	25.0	–	–	36.8	38.0
Both												
Oxford	258	552	526	428	19.5	25.2	16.3	20.0	32.7	35.3	30.7	30.4
England	–	–	6,528	8,207	–	–	16.7	19.8	–	–	30.5	30.3
Sex/Area	UC/M %				Average annual % change			95% CI				
	1979–1983	1984–1992	1993–2000 ^b	2001–2006								
Males												
Oxford	65.3	68.1	47.5***	66.9***	–0.6				–2.1	1.0		
England	–	–	52.7	64.2***	–0.8				–2.5	0.9		
Females												
Oxford	55.2	71.3***	54.5***	64.1*	–0.9†				–1.7	–0.0		
England	–	–	52.8	64.1***	0.6				–0.2	1.3		
Both												
Oxford	58.9	70.1**	52.1***	65.2***	–0.7				–1.5	0.1		
England	–	–	52.7	64.1***	0.0				–0.8	0.9		

* significant change between percentage of mentions coded as underlying cause in the asterisked period, compared with the previous period, at * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ †statistically significant change

^a Age standardised in 5 year age groups, within each of the broader age groups

^b 1996–2006 for England data

34, and 38. The average annual age-standardised mortality rate for women, measured as underlying cause, was 17 in 1979–1983, 19 in 1984–1992, 19 in 1993–2000 and 23 in 2001–2006. Measured as mentions, the corresponding rates in these four time periods were 22, 20, 22 and 26.

Overall, allowing for shifts in coding between mentions and underlying cause, mortality rates for MND in the Oxford region showed no evidence of any sustained change over time. However, there were striking increases in mortality rates in women aged 75 years and over. The average annual age-specific rates for underlying cause of death in this age group were 41 in 1979–1983, 94 in 1984–1992, 103 in 1993–2000 and 120 in 2001–2006. The corresponding rates for mortality, based on mentions, were 70, 108, 118 and 136. Thus, the rise in rates based on underlying cause in this age group cannot be attributed simply to shifts in coding between underlying cause and mentions.

The average annual increase in mortality for mentions in women aged 75 and over was 2.4% (95% confidence intervals 0.7 to 4.2%). There was a small and non-significant rise in mortality rates for men aged 75 and over.

The large English dataset, available for mentions over the much shorter period of time, 1996 to 2006, showed a significant increase in mortality of 2.7% (95% confidence intervals 1.8, 3.6). The annual rates in 1996–2000 rose slightly; there was an abrupt upward shift between 2000 and 2001 coinciding with the change in the International Classification of Diseases editions from ICD9 to ICD10; and after that, there was a continuing small rise in mortality, particularly in most recent years (Fig. 1b). Detailed examination of the data showed that the increases in mortality were most evident in people aged over 75 years: the mentions rate in men 2006 was 11% higher than that in 2001, and that in women was 7% higher.

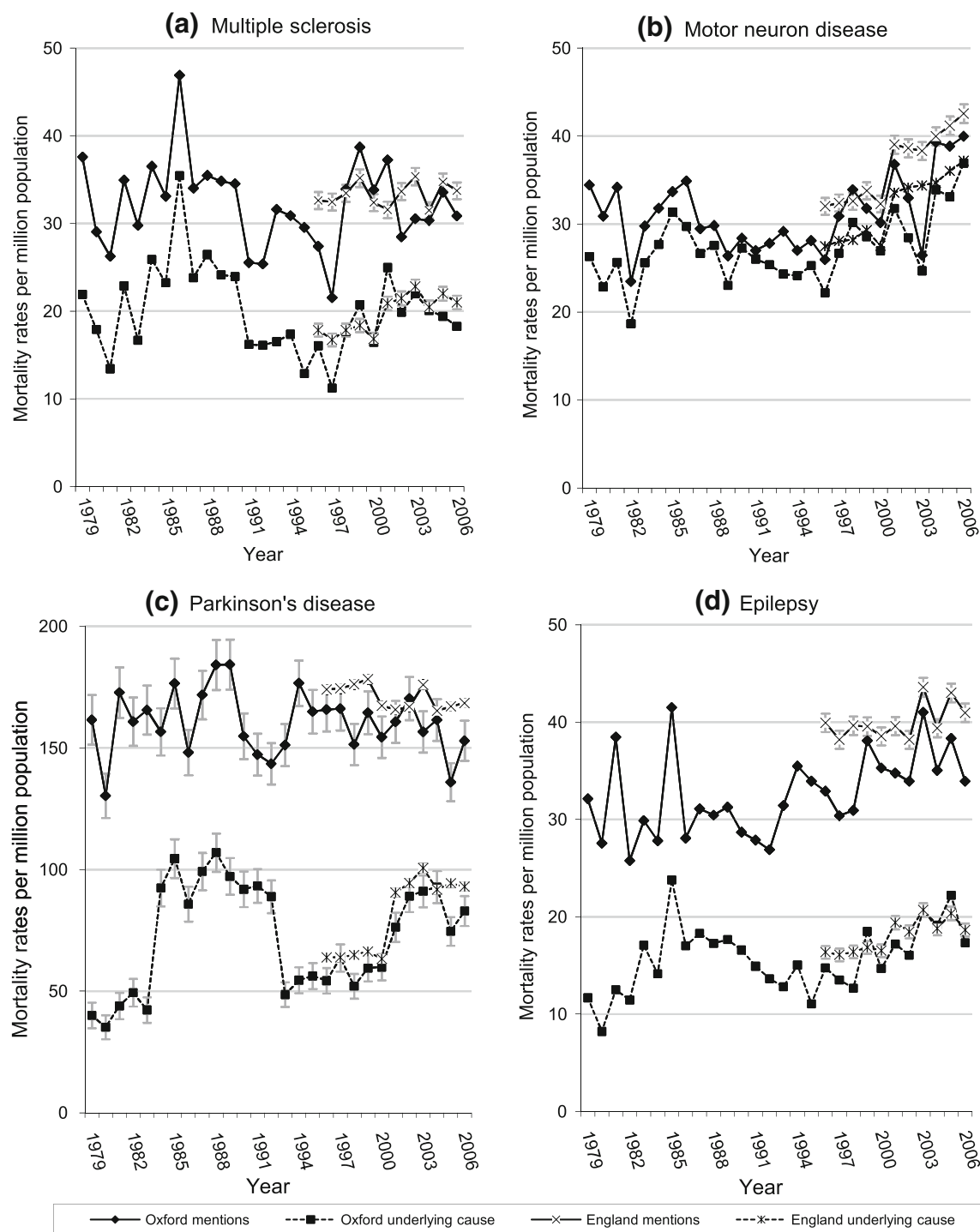


Fig. 1 Age standardised mortality rate for multiple sclerosis, motor neuron disease, Parkinson's disease and epilepsy in the Oxford region 1979–2006 and England 1996–2006, comparing underlying cause and

mentions. (Note that diseases are presented with different scales on the vertical axis)

Parkinson's disease (Table 3; Fig. 1c)

Underlying cause analysis identified 53% of all deaths in Oxford from 2001 to 2006 and 55% of all deaths in England. In the Oxford analysis, underlying cause coded

mortality identified 27% of all certified deaths that mentioned PD in the first period of the study, 59% in the second, 35% in the third and 54% in the fourth (Table 3). The Oxford data for underlying cause (Fig. 1c, lower line) shows the substantial impact of changes to

Table 2 Motor neuron disease: age-standardised death rates in the Oxford Region 1979–2006 and in England 1996–2006: rates per million population; underlying cause deaths as a percentage of all mentions (UC/M %); average annual change in death rates based on mentions

Sex/Area	Number of deaths				Rates ^a per million							
	Mentions				Underlying cause				Mentions			
	1979–1983	1984–1992	1993–2000 ^b	2001–2006	1979–1983	1984–1992	1993–2000 ^b	2001–2006	1979–1983	1984–1992	1993–2000 ^b	2001–2006
Male												
Oxford	149	287	281	266	31.4	34.8	30.3	33.2	40.3	38.5	34.4	37.9
England	–	–	3,462	5,460	–	–	28.2	34.5	–	–	33.0	39.7
Female												
Oxford	101	201	223	215	16.8	18.8	19.2	23.4	21.8	20.2	21.5	25.9
England	–	–	2,947	4,350	–	–	19.3	22.7	–	–	21.5	25.0
Both												
Oxford	250	488	504	481	23.3	25.9	24.5	27.8	29.7	28.2	27.5	31.3
England	–	–	6,409	9,810	–	–	23.3	28.0	–	–	26.5	31.6
Sex/Area	UC/M %				Average annual% change				95% CI			
	1979–1983	1984–1992	1993–2000 ^b	2001–2006								
Male												
Oxford	79.9	90.2**	87.5	87.2	–0.6				–1.2	0.0		
England	–	–	84.9	86.0	2.9†				1.8	4.1		
Female												
Oxford	75.2	92***	88.3	89.3	0.7				–0.4	1.8		
England	–	–	87.7	89.6*	2.2†				1.2	3.2		
Both												
Oxford	78.0	91.0***	87.9	88.1	0.1				–0.5	0.7		
England	–	–	86.2	87.6**	2.7†				1.8	3.6		

* significant change between percentage of mentions coded as underlying cause in the asterisked period, compared with the previous period, at

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ †statistically significant change

^a Age standardised in 5 year age groups, within each of the broader age groups

^b 1996–2006 for England data

coding rules on the apparent trends in mortality from PD.

Trends based on mentions show a significant but small fall in the death rate in Oxford over the time period from 1979 to 2006 (Table 3). The largest decreases were in people aged 55–64 years (data not shown). National trends also show a significant fall over a much shorter period. Overall, mortality rates for men were approximately double those for women and, in both sexes, they rose very steeply with age. As Table 3 and the figure show, the rates for the Oxford region were a little lower than those for England but they were broadly similar.

Epilepsy (Table 4; Fig. 1d)

From 2001 to 2006, 52% of deaths in Oxford and 48% in England with epilepsy mentioned on death certificates were recorded with epilepsy as the underlying cause. Like the pattern for MS and PD, the percentage of deaths from

epilepsy in Oxford that were coded as the underlying cause rose significantly between 1979–1983 and 1984–1992, fell significantly between 1984–1992 and 1993–2000, and then rose significantly again between 2001 and 2006 (Table 4). This represents the effect of changes to the selection of the underlying cause of death. Mortality was generally higher for men than women, most notably in the age range 35–54, where mortality was almost twice as high in men (data not shown). Overall, mortality did not increase significantly in Oxford between 1979 and 2006 or in England between 1996 and 2006. In Oxford, albeit based on small numbers, there was a significant upward trend in mentions-based mortality from epilepsy in males aged under 15 years between 1979 and 2006: the rate rose from 6.1 to 12.6 per million males aged under 15 from 1979–1983 to 2001–2006 (average annual change 8.1%, 95% confidence interval 0.1 to 17.3). There was a similar rise in females under 15 (from 8.1 to 11.6) that did not reach statistical significance.

Table 3 Parkinson's disease: age-standardised rates in the Oxford Region 1979–2006 and in England 1996–2006: rates per million population; underlying cause deaths as a percentage of all mentions (UC/M %); average annual change based on mentions

Sex/Area	Number of deaths				Rates ^a per million							
	Mentions				Underlying cause				Mentions			
	1979–1983	1984–1992	1993–2000 ^b	2001–2006	1979–1983	1984–1992	1993–2000 ^b	2001–2006	1979–1983	1984–1992	1993–2000 ^b	2001–2006
Males												
Oxford	656	1,415	1,520	1,244	50.6	113.3	58.3	87.6	198.1	193.4	178.0	164.8
England	–	–	19,222	23,954	–	–	60.1	83.8	–	–	164.9	152.3
Females												
Oxford	641	1,244	1,230	858	29.1	53.6	30.2	37.8	103.7	91.3	80.7	68.0
England	–	–	14,971	17,332	–	–	26.2	36.8	–	–	69.4	64.5
Both												
Oxford	1,297	2,659	2,750	2,102	36.3	74.4	40.2	56.7	136.1	126.7	115.6	105.3
England	–	–	34,193	41,286	–	–	38.6	54.6	–	–	103.6	97.9
Sex/Area	UC/M %				Average annual % change				95% CI			
	1979–1983	1984–1992	1993–2000 ^b	2001–2006								
Males												
Oxford	26.2	59.0***	33.0***	52.9***	–0.9†				–1.3	–0.5		
England	–	–	36.5	55.0***	–1.3†				–1.8	–0.9		
Females												
Oxford	27.1	58.4***	36.7***	55.7***	–1.9†				–2.4	–1.3		
England	–	–	37.7	57.3***	–1.2†				–1.9	–0.5		
Both												
Oxford	26.7	58.7***	34.7***	54.0***	–1.2†				–1.6	–0.8		
England	–	–	37.0	55.9***	–0.9†				1.5	–0.4		

* significant change between percentage of mentions coded as underlying cause in the asterisked period, compared with the previous period, at *** $p < 0.001$ †statistically significant change

^a Age standardised in 5 year age groups, within each of the broader age groups

^b 1996–2006 for England data

Discussion

Mortality statistics based on the coded underlying cause of death substantially underestimate deaths with MD, PD or epilepsy on the death certificate. They identify the great majority of deaths with MND. In studying trends over time, incompleteness may not matter in itself; as long as the level of incompleteness is consistent over time, it may reveal trends reliably. However, as we show, there have been important changes over time in the rules for the selection of underlying cause; and these mean that statistics based on underlying cause alone are not comparable over time. The rule changes mainly concern the way that 'rule 3' in the ICD is applied [31]. This rule covers the selection of the underlying cause of death when diseases like pneumonia or heart failure, which can be regarded as modes of dying rather than causes of death, are certified as the underlying cause on the same certificate as other diseases (in the

present context, alongside MS, MND, PD or epilepsy). The instructions in 'rule 3' advise the coders of mortality statistics that they should generally select and code the 'other' disease (i.e. not to select pneumonia, heart failure, etc.) as the underlying cause of death. The rule has, however, been applied with different levels of rigour at different times in England. For example, the change in 1984 reduced the selection of diseases like pneumonia as the underlying cause, with a reciprocal increase in the selection of deaths certified as (for example) PD as the underlying cause, when the latter was on the same certificate as pneumonia. The change in 1993 largely reversed this practice.

It is also important to recognise that international comparisons of mortality rates based on underlying cause for neurological diseases—particularly for MS, PD and epilepsy—will be influenced by national coding conventions, the application of rule 3, and the times of change to selection and coding rules.

Table 4 Epilepsy: age-standardised death rates in the Oxford Region 1979–2006 and in England 1996–2006: rates per million population; underlying cause deaths as a percentage of all mentions (UC/M %); average annual change in death rates based on mentions

Sex/Area	Number of deaths				Rates ^a per million							
	Mentions				Underlying cause				Mentions			
	1979–1983	1984–1992	1993–2000 ^b	2001–2006	1979–1983	1984–1992	1993–2000 ^b	2001–2006	1979–1983	1984–1992	1993–2000 ^b	2001–2006
Males												
Oxford	164	326	423	325	13.4	18.5	17.3	20.0	35.5	32.9	39.8	38.0
England	–	–	5,175	6,464	–	–	19.7	21.3	–	–	40.5	39.7
Females												
Oxford	156	291	289	276	11.6	13.4	9.7	14.1	25.3	23.4	21.6	25.6
England	–	–	4,362	5,785	–	–	11.4	13.4	–	–	25.5	27.3
Both												
Oxford	320	617	712	601	12.4	16.0	13.5	17.1	29.4	27.9	30.1	31.7
England	–	–	9,537	12,249	–	–	15.5	17.3	–	–	32.5	33.2
Sex/Area	UC/M %				Average annual % change				95% CI			
	1979–1983	1984–1992	1993–2000 ^b	2001–2006								
Males												
Oxford	39.6	57.7***	43.7***	52.3*	0.7				–0.3 1.7			
England	–	–	46.4	51.3***	–0.4				–1.3 0.6			
Females												
Oxford	39.7	54.0**	39.8***	51.4**	–0.1				–1.2 1.1			
England	–	–	36.7	43.4***	1.1				–0.1 2.2			
Both												
Oxford	39.7	55.9***	42.1***	51.9***	0.4				–0.2 0.9			
England	–	–	42.0	47.6***	0.3				–0.5 1.1			

* significant change between percentage of mentions coded as underlying cause in the asterisked period, compared with the previous period, at

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ †statistically significant change

^a Age standardised in 5 year age groups, within each of the broader age groups

^b 1996–2006 for England data

Across all the diseases studied, there has also been a change in healthcare practice over the last three decades in England. These will have resulted in increased referral rates for specialist diagnoses, particularly among more elderly groups, and increased availability of modern diagnostic investigations e.g. neuroimaging. Such changes potentially confound any apparent change in mortality in the elderly, who may not have been as accurately diagnosed in preceding decades.

Multiple sclerosis

A decline in mortality from MS in the UK has previously been reported across 30 years up to 1983 [30]. This contrasts with a recent study of US national mortality data which found increased mortality between 1990 and 2001 [23]. In other countries, such as Canada, rates have been reported to fluctuate over time rather than show a

directional trend [29]. We found no evidence of either a rise or decline in the Oxford region overall, from 1979 to 2006. Our data (Fig. 1a) suggest that there may have been some fluctuations over fairly long timescales: a gradual rise in mention-based rates from the late 1970s to the mid-1990s, a fall from then to the late 1990s, and then a rise. Over the relatively long period we have studied, we conclude from mention-based mortality data that there has been no overall long-term change in mortality from MS.

Motor neuron disease

The diagnostic accuracy of MND based on death certificates is generally considered to be good. A study in Scotland estimated it to have a positive predictive value of 90%, and a false negative rate of 6% [4].

Our finding of an apparent recent increase, and the literature on trends in MND, needs comment. Most

publications on trends over time have shown an increase in recent decades, but with some inconsistencies in respect of geography and age-sex groups. A recent review of 1990s' data from European and North American studies showed an increase in mortality for MND in the elderly and in women, but it also showed wide geographical variation in the extent of any increase [33]. National mortality data for the US from 1969 to 1998 also showed a rise in mortality which was more pronounced in women; and, in that study, during the last decade of the study period (1988–1998) rates among men remained stable while rates in women continued to rise [17]. These authors and others also provided evidence of geographic differences in the death rate associated with MND within the US, with most northern states experiencing higher death rates than southern states [25]. A study of mortality in Great Britain for the period 1975–2004 reported that rates in people under 60 years of age had remained stable, but that there had been an average annual increase in the over 60s of between 4% (females in Scotland) and 6% (males in Scotland) [7]. It is unclear whether the authors were able to take account of changes between coding of MND as the underlying cause or as mentions.

An earlier study from the Oxford region, covering the period 1963–1985, with validation of hospital statistics and mortality data against clinical case notes, suggested that there had been a modest increase in admission and mortality rates for MND that was more convincing in women than in men [8]. Our current findings in Oxford do not provide evidence of any sustained change in mortality over time, since then, in this English population among people aged under 75 years. We did, however, like others, find an increase in mortality rates among women aged 75 and over. Part of this observation may be explained by increased life expectancy in this group, given that the incidence of the disease continues to rise with age in those over 75. The increasing incidence with increasing age, and the generally higher life expectancy in women than men, is reflected in the gradual narrowing of the male predominance of MND noted with each increased decade of age [5]. It is also necessary to consider the potential confounding effect of increased specialist referral of elderly patients and more accurate diagnosis, resulting from use of modern diagnostic techniques e.g. neuroimaging in recent decades.

Chancellor et al. [5], like others before and after [16, 26], observed an excess of bulbar-onset MND in elderly women. In general, the bulbar-onset phenotype of MND is associated with a poorer prognosis compared to limb-onset. However, since MND has a poor prognosis for all its phenotypes, it seems unlikely that any changes in phenotype alone would contribute to a sustained change in mortality. Interestingly, a previous cohort analysis of MND in England and Wales (considering deaths from 1940 to

1982) showed that age-specific mortality rates in women actually declined with age after the age of 65 years [15]. The Scottish ALS/MND Register study over the period 1989–1998 identified an apparent decline in survival, with gender seemingly unrelated to survival [9].

Changes to the nomenclature and classification of diseases, in different revisions of the ICD, probably account for the stepwise increase in MND, both as underlying cause and as mentions, comparing rates before and immediately after 2001. The stepwise change coincided with the introduction in England of the tenth revision of the ICD.

Population based mortality rates are a product of the incidence of the disease and survival. No overall change in a death rate could result, in principle, from an increase in incidence coupled with an increase in survival. This is theoretically possible for a heavily age-dependent disease like MND within an ageing population, coupled with advances in the use of multi-disciplinary patient care [27], and (less plausibly given its very modest effects) the introduction in 1995 of the only licensed disease-modifying treatment to date—riluzole [28].

Parkinson's disease

Rule changes had a substantial impact on the trends in underlying cause-coded mortality for PD. Analysis of mentions, which obviates the need to decide which of several co-morbidities is the underlying cause, showed that there was a modest decline over time in mortality ascribed to PD. This has also been reported in England by others, over a shorter period of time than here [13]. As for the other diseases, improved diagnostic accuracy over recent years must be considered in interpreting the decline of PD, with fewer patients erroneously labelled as PD, and with increased referral of elderly patients for specialist opinion and diagnosis.

Like the present study, previous studies have shown that only about half of certified deaths with PD have PD coded as the underlying cause of death [10, 22].

The view that neurodegenerative conditions such as PD, Alzheimer's disease and MND might share a small number of common genetic and/or environmental risk factors [2] is not supported by our finding that death rates for PD are declining, death rates for certified Alzheimer's disease are increasing [12], and MND mortality appears generally stable throughout most of the period studied by us, except for a recent modest increase.

Epilepsy

It is recognised that death certificate data do not provide full ascertainment of mortality associated with epilepsy [1, 3, 6, 18]. The National Sentinel Clinical Audit of

Epilepsy-Related Death concluded that it was difficult to establish the true number of epilepsy-related deaths from national data and that action should be considered to improve the quality of death certification in relation to these deaths [14]. Our study shows that when epilepsy is mentioned on a death certificate it is recorded as the underlying cause of death in less than half of all cases. Our findings for England for 1996–2006, and the findings of others for different, recent time periods [1, 14], indicate that epilepsy mortality has probably been stable in recent years.

Conclusion

Studies of these neurological diseases based solely on underlying cause of death, underestimate the true ‘burden’ of fatal disease, notably in epilepsy and multiple sclerosis. Studies of trends, based on underlying cause alone, can be misleading because of changes to the rules for selecting and coding the underlying cause. Studies of mortality from chronic neurological conditions should include analysis of all certified causes of death, as well as those selected and coded as the underlying cause.

Permission for the programme of work to analyse these datasets was given by the Central and South Bristol Research Ethics Committee (04/Q2006/176) in its role as a Multi-Centre Research Ethics Committee.

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Conflict of interest statement None.

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